

MBI801: Information Systems Analysis and Design
Assessment One: Information System Project Proposal

Project Title: Pharmacy management system

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Task 1: System vision document (by Claire)

1.1 Problem Description

Many small and medium-sized pharmacies just rely on manual processes for prescriptions, inventory and customer information. This mode led to several significant challenges:

- Inefficient prescription management (Karimi et al., 2025) → customer needs to deliver their prescriptions in person
- Limited inventory control (Jaju et al., 2023) → medicine stocks often face shortages and overstocking issues
- Poor customer tracking (Al-Jumaili et al., 2020) → customer purchase history and medicine usage are lost and further cause the difficult to manage long-term treatment plans
- Payment and order processing delays due to processed manually (Yulia et al., 2025) → checkout times extend, and the risk of error rates increased.

As a result, pharmacies are facing lower operational efficiency, rising operational costs, higher risks of medication errors and declining customer satisfaction and experience. To address these issues, an integrated pharmacy management system is required.

1.2 System capabilities

- 1) Prescription upload and review: the system allows customers to upload prescriptions online and enables pharmacists to review, approve, or reject them directly.**
 - Secure online prescription upload
 - Pharmacist approval or rejection
 - Automated prescription record management
- 2) Order management and payment: the system supports multi-channel order placement and integrates a secure payment gateway for reliable transactions.**
 - Place orders online or in-store
 - Integrated secure payment gateway
 - Real-time order tracking
- 3) Inventory management and automatic replenishment: the system monitors inventory levels in real time and automatically sends low-stock alerts and supplier notifications.**
 - Real-time inventory monitoring
 - Automatic low-stock alerts

- Supplier replenishment notifications
- 4) **Customer management and medicine tracking: the system maintains customer profiles, tracks purchase history, and manages medicine usage data.**
- Maintain customer profiles
- Track purchase history
- Enable personalized recommendations

1.3 Business benefits

Implementing the pharmacy management system will bring significant following benefits to both pharmacy and its customers:

- Higher efficiency: the system can automatically handle prescription review, order processing and inventory updates, improving overall operational efficiency.
- Better customer experience: pharmacies can improve prescription check time and stock information faster, leading to quicker services and a better customer experience.
- Improve accuracy: the system can validate prescriptions automatically.
- Optimized inventory control: the system can automatically restock medicines, ensuring inventory control and avoiding shortages.
- Better business insights: the system analyzes customer purchase history and medicine usage to improve business planning and provides more personalized recommendations.

1.4 Stakeholder Analysis

Stakeholder Identification

1) Primary Stakeholders:

- Customers/ Patients
- Pharmacists
- Pharmacy Managers
- Suppliers
- Delivery Personnel

2) Secondary Stakeholders:

- Health Inspectors
- IT Support/ System Administrators

- Healthcare Regulators
- Insurance Companies
- Data Analysts

3) External Stakeholders:

- Drug Manufactures
- Delivery Companies
- Government Agencies
- API Vendors

Stakeholder Classification:

By Organizational Level:

Operational: Pharmacists, Customers/Patients, Delivery Personnel, Receptionist, IT Support/ System Administrators, Inventory clerk, Customer Support, Suppliers, API vendors

Management: Pharmacy Managers, Health Inspectors, Accounting Manager

Executive: Health Policy Maker, Government Regulator, Pharmacy Chain Director, Insurance Provider Executive

By Location:

- Internal: Pharmacists, Inventory clerk, IT Support/ System Administrators, Customer Support, Pharmacy Managers, Quality Assurance Officer, Accounting Manager, Pharmacy Chain Director, Health Inspectors
- External: Customers/Patients, Delivery Personnel, Receptionist, Health Policy Maker, Government Regulator, Suppliers, Insurance Provider Executive, API vendors

Stakeholder	Organizational Level	Internal/External
Customers/Patients	Operational	External
Pharmacists	Operational	Internal
Delivery Personnel	Operational	External
IT Support/ System Administrators	Operational	Internal
Receptionist	Operational	External
Inventory clerk	Operational	Internal
Customer Support	Operational	Internal

Pharmacy Managers	Management	Internal
Suppliers	Operational	External
Health Inspectors	Management	Internal
Accounting Manager	Management	Internal
Health Policy Maker	Executive	External
Government Regulator	Executive	External
Pharmacy Chain Director	Executive	Internal
Insurance Provider Executive	Executive	External
API vendors	Operational	External

1.5 User Goal Identification

Customer / Patient:

- Upload prescription
- Track order status
- Complete payment
- Manager personal profile

Pharmacist:

- Review prescription
- Update prescription status
- Prepare medication

Pharmacy Manager

- Monitor inventory levels
- Assign pharmacist tasks
- Review inventory reports

Suppliers:

- Deliver medication stock
- Receive purchase orders
- Confirm delivery status

Delivery Personnel:

- View delivery schedule
- Confirm delivery completion
- Report delivery issue

Health Inspector

- Review pharmacy compliance
- Generate audit report

Customer Support

- Answer customer questions
- Notify order updates

IT Support/ System Administrators

- Maintain system
- Manager user access
- Resolve technical issues

Pharmacy Chain Director

- Monitor pharmacy performance
- Review regulatory compliance
- Approve system-wide updates
- Analyze sales and prescription trends

Task 2: Event Decomposition (by Claire)

External Events→ Use Cases:

- Customer uploads prescription → Review Prescription
- Customer places a medicine order → Process Order
- Customer completes payment online → Confirm Payment
- Pharmacist updates prescription status → Notify Customer
- Supplier delivers new stock → Update Inventory

- Customer registers or updates profiles → Manage Customer Profile

Temporal Events → Use Cases:

- End of each day → Generate Daily Sales Report
- Weekly inventory check → Check Stock
- Monthly summary deadline → Generate Monthly Summary

State Events → Use Case:

- Medicine stock falls below threshold → Trigger Reorder
- Prescription approved → Prepare Medication
- Payment confirmed → Issue Receipt

Event Type	Triggering Event	Use Case (Verb-Noun)
External	Customer uploads prescription	Review Prescription
External	Customer places a medicine order	Process Order
External	Customer completes payment online	Confirm Payment
External	Pharmacist updates prescription status	Notify Customer
External	Supplier delivers new stock	Update Inventory
External	Customer registers or updates profiles	Manage Customer Profile
Temporal	End of each day	Generate Daily Sales Report
Temporal	Weekly inventory check	Check Stock
Temporal	Monthly summary deadline	Generate Monthly Summary
State	Medicine stock falls below threshold	Trigger Reorder
State	Prescription approved	Prepare Medication
State	Payment confirmed	Issue Receipt

Task 3 Use Case Diagram (by Claire + Umanga)

3.1 Use case list (by Claire)

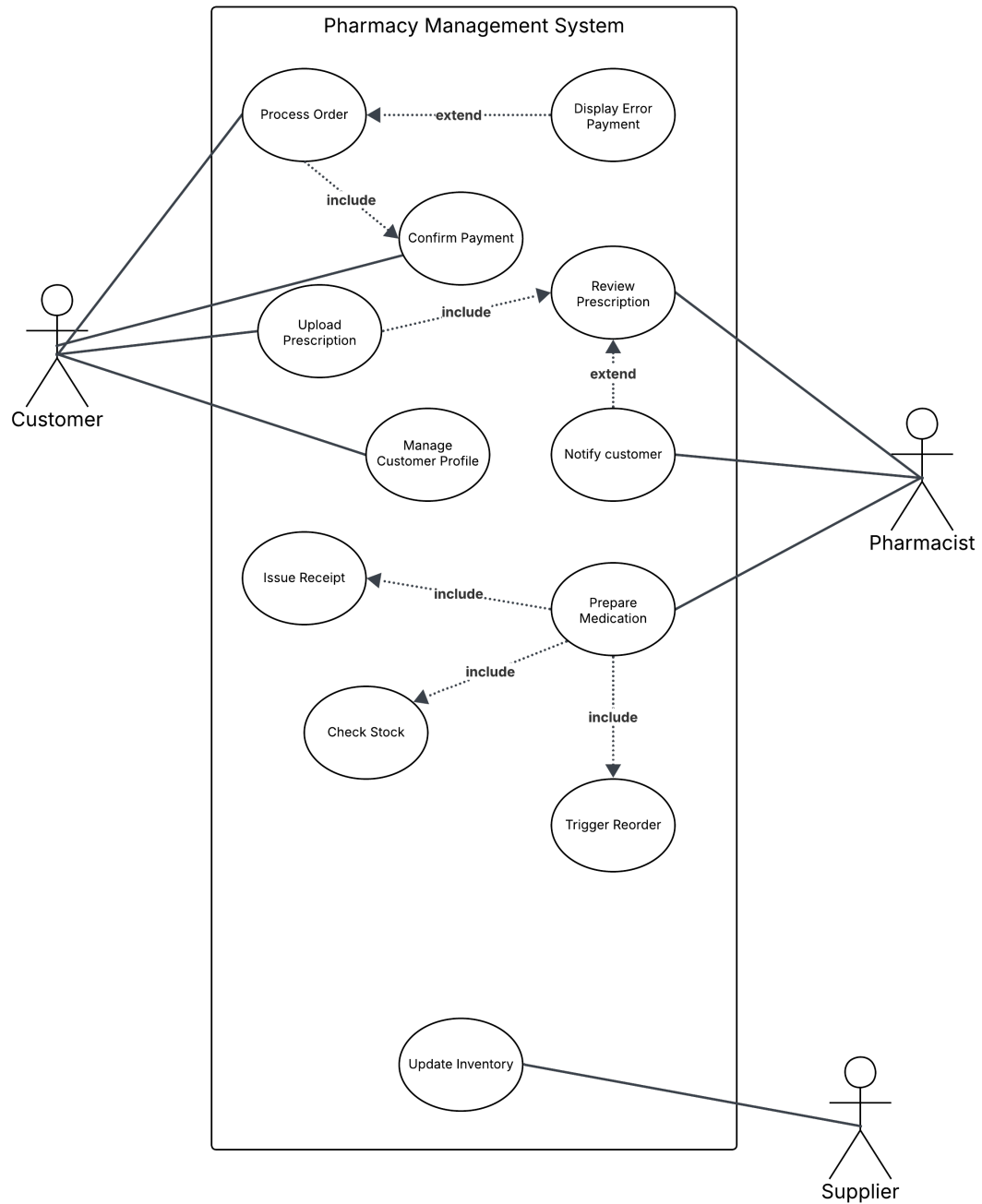
Primary Actor	Use case
Customer	Upload prescription
	Process order
	Confirm payment
	Manage customer profile
Pharmacist	Review prescription
	Notify customer
	Prepare medication
Supplier	Update inventory
System	Check stock
	Trigger recorder
	Issue receipt

Include/ Extend:

Process order ^{Include} **→** **Confirm**
payment ^{Extend} **←** **Display payment**
error

Upload prescription ^{Include} **→** **Review**
prescription ^{Extend} **←** **Notify customer**
Prepare medication ^{Include} **→** **Issue**
receipt

3.2 Use case diagram (by Umanga)



Task 4 Detailed Two Use Case Descriptions (by Abirami)

4.1 Use Case - Upload Prescription

Actor - Customer/Patient

Stakeholder:

- System
- Pharmacist
- Doctor
- Patient

Goal:

The patient must upload a prescription without errors for the pharmacist to validate it successfully.

Preconditions:

- The prescription must be authorised by a healthcare professional.
- The prescription must have clear and concise details of the medication's name.
- The prescription must have the most recent date of issue approved.
- The prescription must contain the correct amount and dosage of medication.
- The customer must be registered into the system.

Steps:

1. The customer logs in.
2. The customer uploads the prescription to the system (JPEG, PDF).
3. The system validates and secures the data.
4. The system sends a notification to the pharmacist.
5. The pharmacist receives the notification about the uploaded prescription.
6. The pharmacist reviews the prescription for authorisation and authenticity, and checks the date it was issued to the patient.

Postconditions:

- Prescription stored in the database.
- Prescription status changed to "Pending Review".
- The pharmacist received a reminder notification.

- The system records the upload time and user ID.

Flow of Activity

Main Scenario:

1. Customer logs in to the online platform
2. Customer enters into the “upload Prescription” section
3. Customer uploads a valid prescription file (JPEG,PDF).
4. The system validates the format and saves the file to the patient’s profile.
5. The system sends notification to the pharmacists

Alternative 1: Invalid file format (DOCX)

1. Customer upload an unsupported file format
2. System rejects the file and displays an error message
3. Customer is reminded to reupload a valid format

Alternative 2: unreadable prescription

1. Prescription is upload but not readable by the system
2. System flags the prescription as “unreadable”
3. Customer receives a request to reupload a clearer version

Exception Conditions:

- Lost internet connection during file upload.
- Upload fails due to a corrupted file.
- Prescription is unauthorised doctor’s signature
- System failure while saving the file.

4.2 Use Case - Preparing Medication

Actor – Pharmacist, Customer/Patient

Stakeholder:

- Pharmacist
- Customer / Patient
- System (Pharmacy system)

- Doctor

Goal:

Prepare medication for the customer/patient without any mistakes .

Preconditions:

- The prescription must be verified and approved.
- The medication must be available in stock.
- The pharmacist must check the expiration date and confirm the proper dosage of the medication.

Steps:

1. The pharmacist checks the system for the medication's availability.
2. The system displays the current stock of the medication.
3. The pharmacist prepares the medication according to the dosage and instructions on the prescription.
4. The system is updated to reflect that the medication is prepared.
5. The customer is notified that their medication is ready for pickup or delivery.

Postconditions:

- The medicine is correctly prepared and labelled.
- The prescription details are updated in the customer's profile and the system.
- The medicine is ready to be received by the customer via delivery or pickup.

Flow of Activity

Main Scenario:

1. Pharmacist logs into the system
2. The pharmacist checks the list of prescriptions marked as “ Pending Review”.
3. Pharmacist selects a prescription and reviews the details
4. System displays complete prescription information
5. Pharmacist confirms availability of the medication in stock
6. Pharmacist prepares the required medication
7. System updates prescription status to “Ready to Pickup”
8. Customers are notified via email or message.

Alternative 1: Medication out of stock

1. pharmacist identifies that medication is not available in inventory
2. System provides option to request order from supplier
3. pharmacist contacts the customer for alternatives
4. Prescription status become "Pending with awaiting stock"

Alternative 2: Prescription information unclear

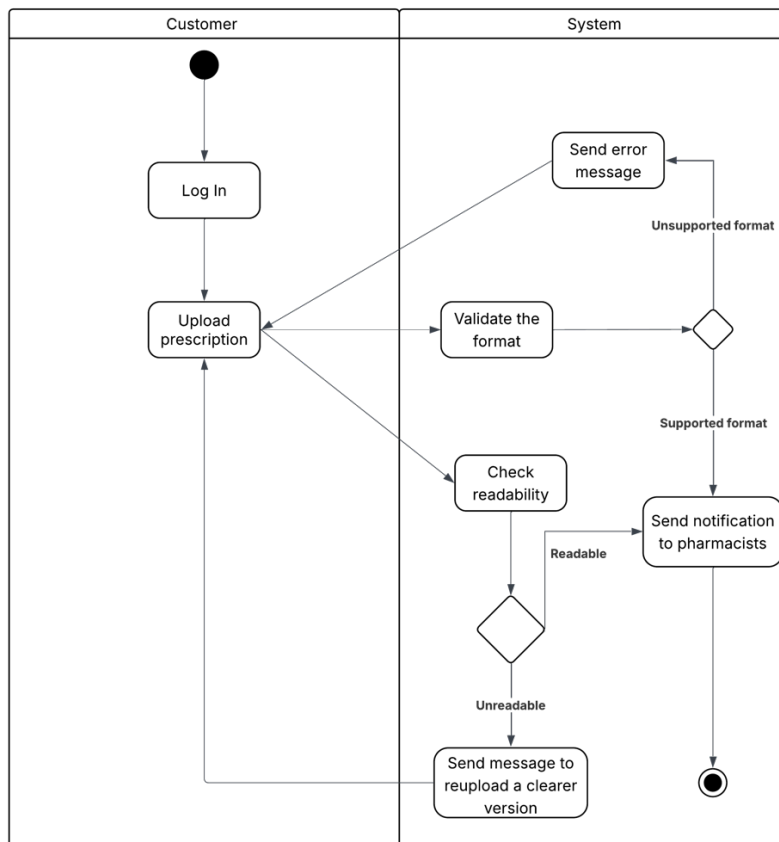
1. pharmacist finds ambiguous details
2. pharmacist contacts the prescribing doctor for clarification
3. System changes into "Hold" status
4. pharmacist keep preparation once information is confirmed

Exception Conditions:

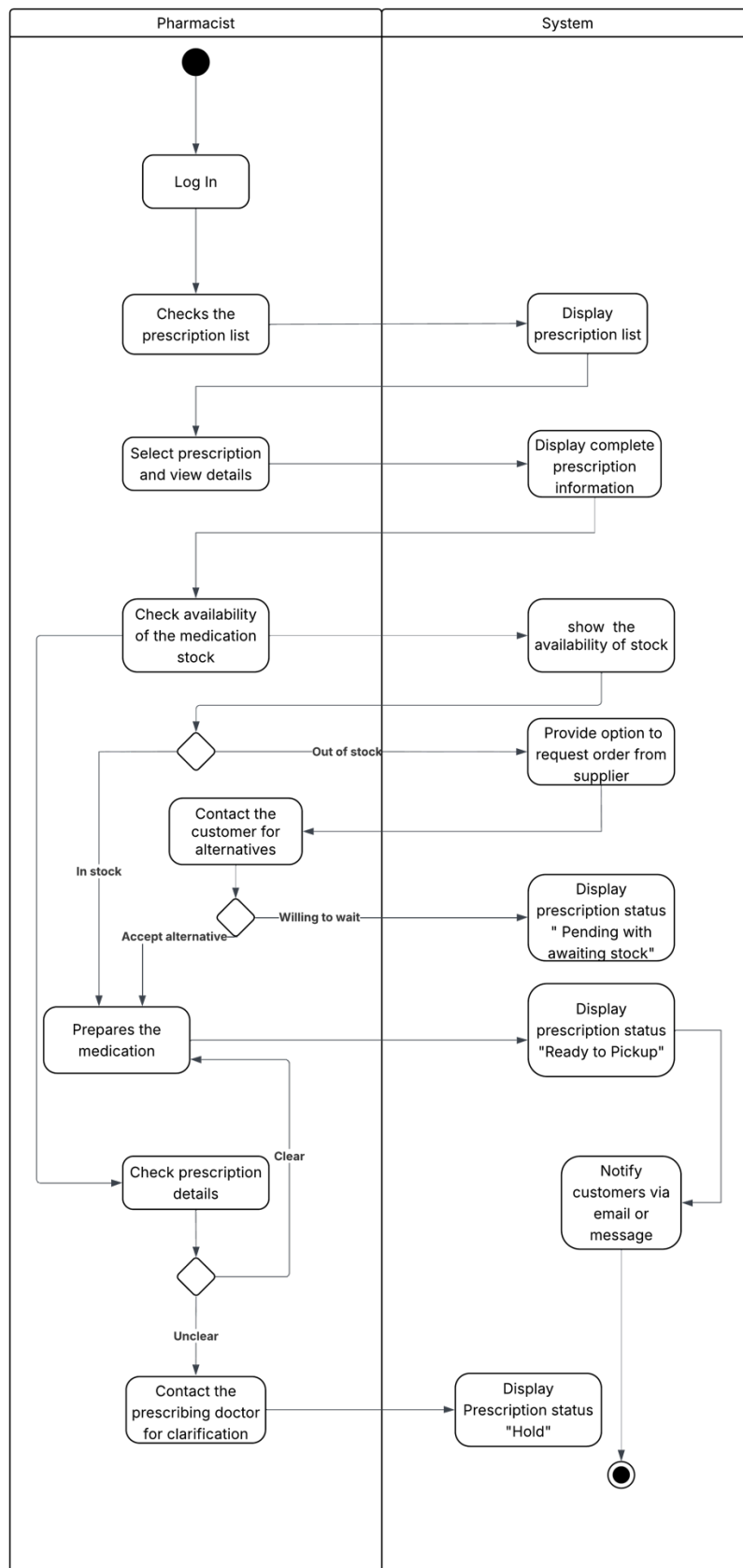
- System crash during prescription access
- Network failure during preparation update
- Prescription is regarded as forged or expired.

Task 5 Two Activity Diagrams (by Umanga)

5.1 Upload Prescription

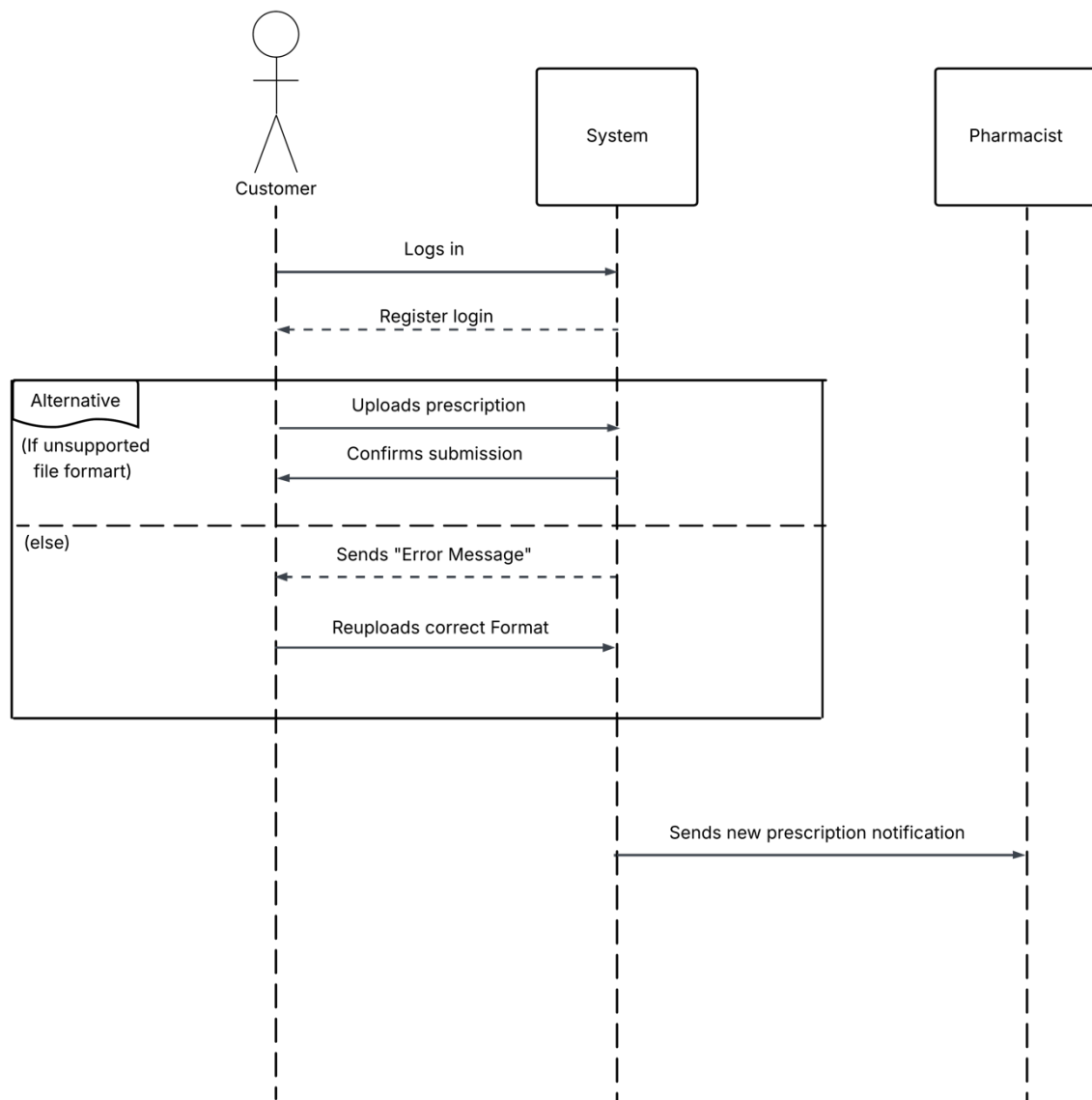


5.2 Preparing Medication

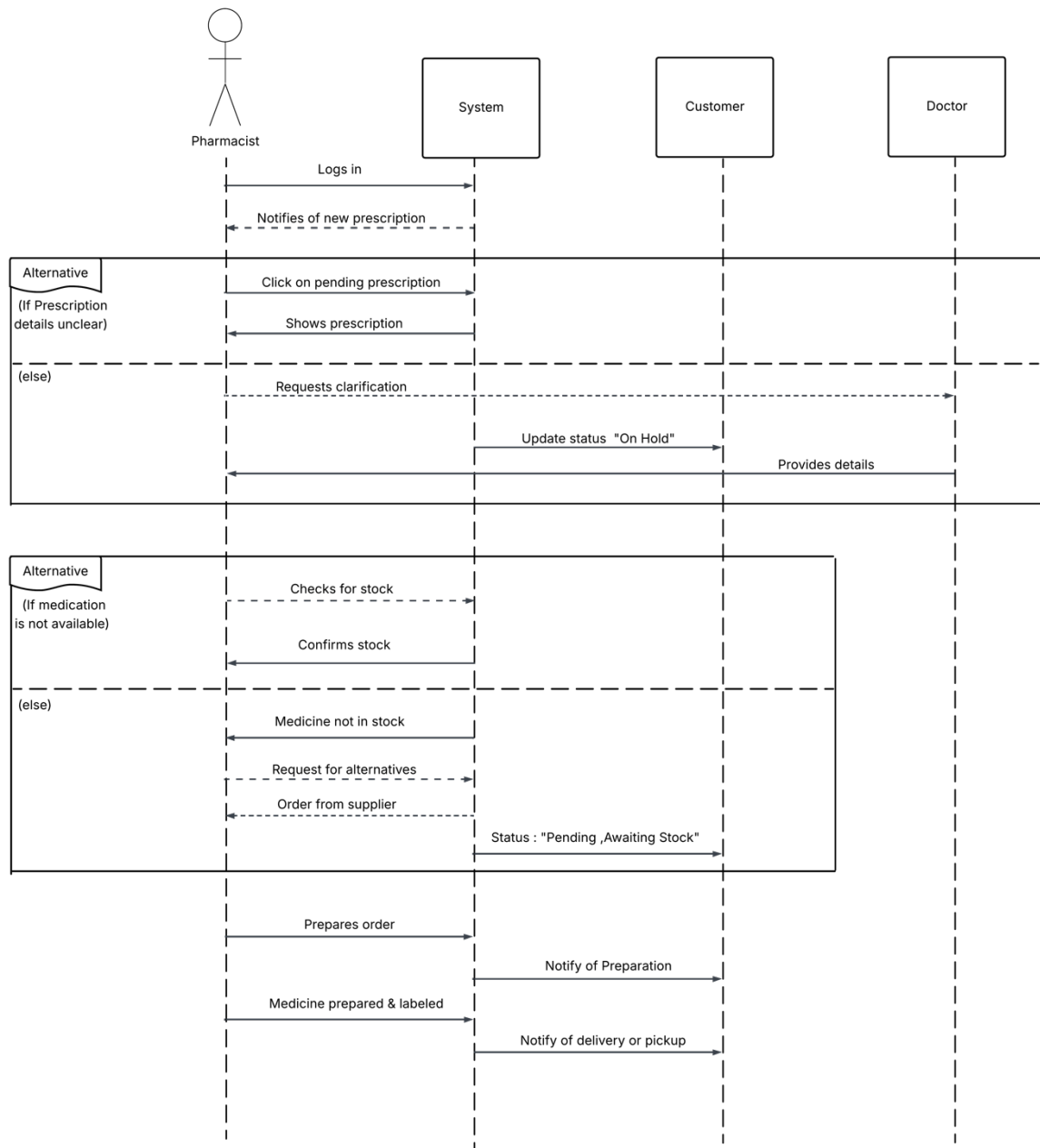


Task 6 Two System Sequence Diagrams (by Abirami)

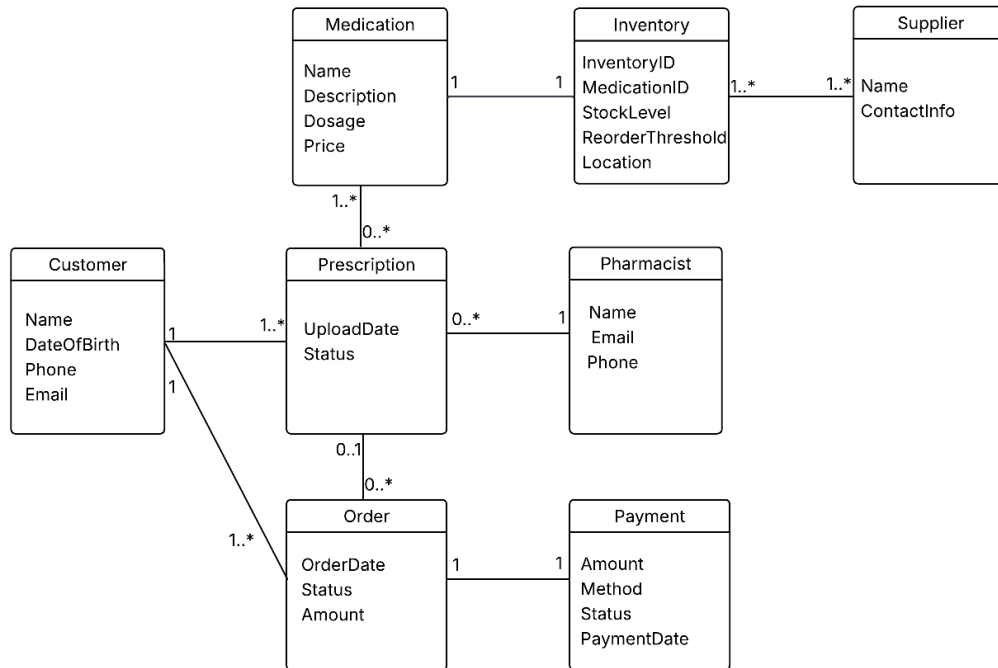
6.1 Upload Prescription



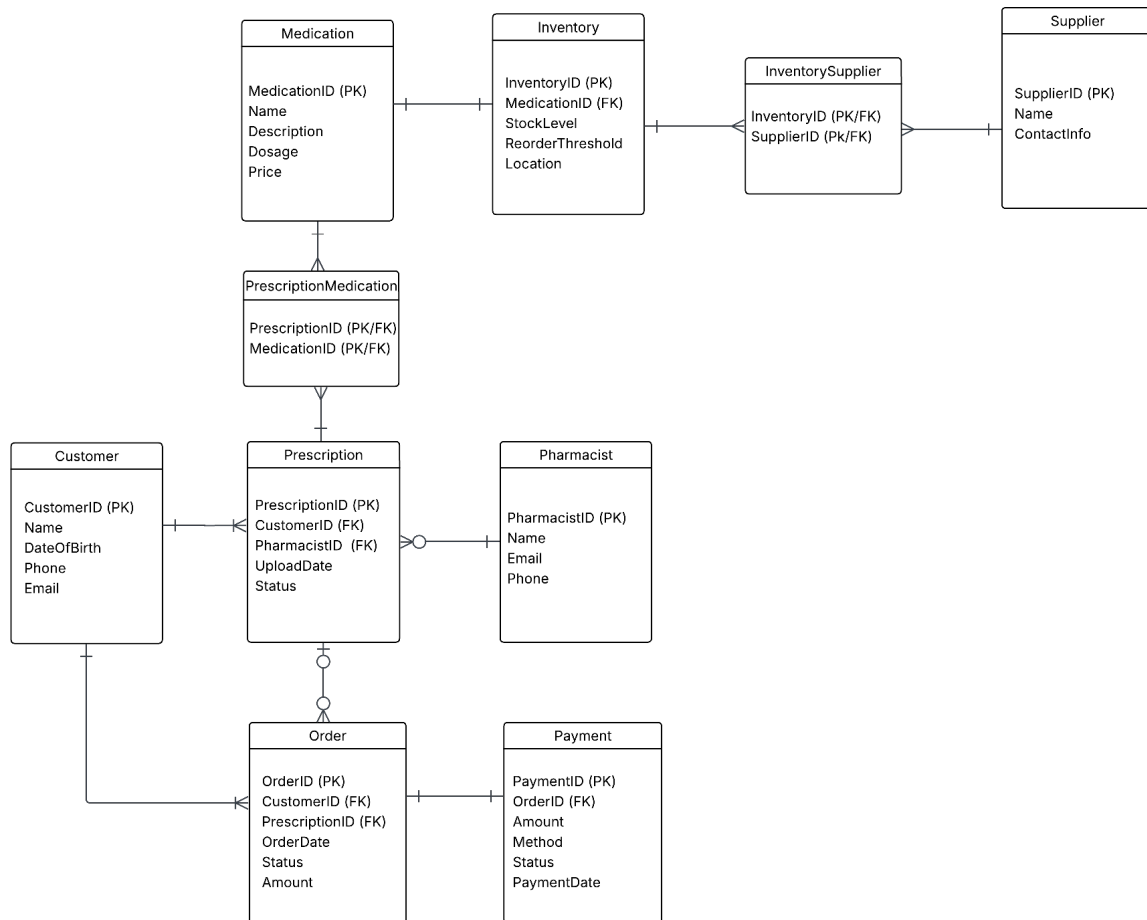
6.2 Preparing Medication



Task 7 Domain Model Class Diagram (by Bryan)



Task 8 ER Diagram (by Bryan)



References

- Al-Jumaili, A. A., Ameen, I. A., & Alzubaidy, D. A. (2020). Influence of Pharmacy Characteristics and Customer Quality of Life on Satisfaction of Community Pharmacy Customers. *Innov Pharm*, 11(1). <https://doi.org/10.24926/iip.v11i1.2434>
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- Karimi, S., Jahani, Y., & Ahmadian, L. (2025). The effect of electronic prescription systems on pharmacy performance through evaluation of existing infrastructure in Kerman Iran. *Sci Rep*, 15(1), 27350. <https://doi.org/10.1038/s41598-025-12629-9>
- Yulia, R., Hartono, R., Indrayanti, M., Ayumuyas, N. P., & Herawati, F. (2025). Studying waiting time in pharmacy: A strategy for improving patient satisfaction. *MethodsX*, 14, 103282. <https://doi.org/10.1016/j.mex.2025.103282>

Time and Task assignment

Main Task assignment

Member	Task
Claire Li	System Vision +Event decomposition +Use case list
Umanga	Use case diagram +Activity diagrams
Abirami	Detailed use case descriptions (Supported by Claire)+System sequence diagrams
Bryan	Domain model class diagram +ER diagram

Timetable

Task No.	Task Name	Major leader	Predictable Date	Completed time	Check by other members
-	Choose project topic	All	06/09/2025	06/09/2025	✓
-	First discussion meeting (in person)	-	07/09/2025 (after class)	07/09/2025 (1:30pm-3:30pm)	-
Task 1	System vision document	Claire	07/09/2025	07/09/2025	✓
Task 2	Event decomposition	Claire	08/09/2025-09/09/2025	09/09/2025	✓
Task 3	Use case list	Claire	09/09/2025	09/09/2025	✓
	Use case diagram	Umanga	10/09/2025	10/09/2025	✓
Task 5	Select 2 core use cases	All	11/09/2025	11/09/2025	✓
	Detailed 2 use case descriptions	Abirami	11/09/2025-12/09/2025	13/09/2025	✓
-	Second discussion meeting (online)	-	12/09/2025	12/09/2025	-

			(after class)	(1:30pm-2:30pm)	
Task 4	2 activity diagrams	Umanga	13/09/2025-14/09/2025	14/09/2025	✓
Task 6	2 System Sequence diagrams	Abirami	13/09/2025-14/09/2025	15/09/2025	✓
-	Third discussion meeting (in person)	-	16/09/2025	16/09/2025 (2:30pm-4:30pm)	-
Task 7	Domain model class diagram	Bryan	15/09/2025-17/09/2025	17/09/2025	✓
Task 8	ER diagram				
-	Merge all sections into one pdf+ Adjust style	Claire	18/09/2025-20/09/2025	18/09/2025	✓
-	Check and Revised	All	21/09/2025	18/09/2025-19/09/2025	✓
-	Submit	All	21/09/2025	19/09/2025-21/09/2025	✓